

HOBURG FYR, Sweden

WMO: 026790

Lat: 56.9209N

Lon: 18.1506E

Elev: 128

StdP: 14.63

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	19.5	22.3	8.2	8.4	24.1	12.0	10.2	25.0	35.3	34.4	32.7	35.1	11.2	20	0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.7	75.6	65.0	73.0	63.6	70.7	62.9	67.9	72.1	66.2	70.6	64.6	68.7	10.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.4	97.8	70.1	64.5	91.3	68.3	62.8	86.0	67.1	32.3	72.3	30.9	70.7	29.8	68.8	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.9	27.5	24.9	17.1	78.9	4.6	4.0	13.8	81.8	11.1	84.2	8.5	86.5	5.2	89.4
			15.0	69.2	4.0	2.8	12.2	71.2	9.9	72.9	7.7	74.5	4.8	76.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.7	33.1	32.4	34.9	41.3	49.4	57.4	63.6	64.2	57.1	48.4	41.3	36.3	(6)
	DBStd	12.23	5.13	5.37	4.73	4.83	5.25	4.15	4.53	4.07	4.29	4.76	4.71	5.31	(7)
	HDD50	2600	525	494	469	264	74	1	0	0	2	86	260	425	(8)
	HDD65	6761	990	914	934	710	484	230	83	64	238	514	710	890	(9)
	CDD50	1400	0	0	0	4	56	223	422	439	217	38	1	0	(10)
	CDD65	80	0	0	0	0	0	1	39	38	2	0	0	0	(11)
	CDH74	155	0	0	0	0	1	2	100	51	1	0	0	0	(12)
	CDH80	7	0	0	0	0	0	0	6	1	0	0	0	0	(13)
(14) Wind	WSAvg	13.0	15.2	14.0	13.1	12.0	11.3	11.4	10.9	11.4	13.0	14.1	14.4	15.8	(14)
(15) Precipitation	PrecAvg	20.0	1.6	1.3	1.2	1.1	1.3	1.4	1.9	2.0	2.2	1.9	2.2	1.9	(15)
	PrecMax	28.0	4.8	3.0	3.0	3.3	4.4	3.8	5.7	4.6	5.1	5.0	4.3	3.5	(16)
	PrecMin	14.8	0.2	0.3	0.1	0.1	0.3	0.0	0.0	0.2	0.2	0.2	0.5	0.5	(17)
	PrecStd	3.3	1.0	0.6	0.7	0.7	0.9	0.9	1.4	1.1	1.2	1.1	0.8	0.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.6	42.6	48.0	61.5	69.0	72.7	80.1	78.3	71.1	59.5	52.2	46.7	(19)
		MCWB	42.2	39.7	41.6	49.9	57.7	59.6	66.7	66.7	61.8	56.3	50.5	45.3	(20)
	2%	DB	42.0	40.9	44.1	55.5	64.3	69.7	76.8	75.1	67.2	57.7	50.2	45.6	(21)
		MCWB	40.9	39.2	40.2	46.2	54.9	59.4	66.1	64.6	60.8	55.1	48.8	44.3	(22)
	5%	DB	40.8	39.8	42.1	51.6	60.7	66.8	74.0	72.8	64.9	56.1	48.6	44.3	(23)
		MCWB	39.6	38.3	39.2	44.5	52.9	58.4	64.6	63.7	59.8	53.8	47.3	43.2	(24)
	10%	DB	39.6	38.8	40.5	48.6	57.6	64.5	71.4	70.6	63.1	54.7	47.3	42.8	(25)
		MCWB	38.5	37.4	38.2	43.4	51.2	57.5	63.7	63.4	58.7	52.6	46.0	41.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.9	40.9	43.1	51.1	60.5	64.0	71.1	71.3	64.7	57.5	51.3	46.2	(27)
		MCDB	43.2	41.5	46.3	59.8	66.7	69.3	75.2	75.6	68.1	58.7	51.6	46.4	(28)
	2%	WB	41.3	39.7	41.2	47.6	56.8	61.7	68.4	67.8	62.5	55.8	49.1	44.6	(29)
		MCDB	41.8	40.4	43.2	52.9	61.7	66.7	73.2	71.2	65.4	57.0	49.8	45.1	(30)
	5%	WB	39.9	38.6	39.9	45.6	54.4	60.1	66.8	66.2	61.1	54.5	47.6	43.2	(31)
		MCDB	40.5	39.4	41.5	50.3	59.0	64.6	71.7	70.0	63.8	55.7	48.3	43.9	(32)
	10%	WB	38.7	37.7	38.7	43.8	52.3	58.6	65.1	64.8	59.8	53.1	46.3	41.7	(33)
		MCDB	39.3	38.6	40.1	47.8	56.6	62.8	69.8	68.6	62.3	54.4	47.0	42.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	4.9	6.6	9.6	10.4	9.7	9.2	8.7	7.4	5.5	4.5	4.3	(35)
		MCDBR	4.5	5.3	8.3	14.7	15.1	13.4	13.1	11.1	8.7	6.0	5.0	4.1	(36)
	5% WB	MCWBR	4.7	4.9	5.9	8.0	8.6	6.6	6.1	6.2	5.7	5.0	5.2	4.7	(37)
		MCWBR	4.3	4.8	7.5	12.9	13.3	11.1	11.0	9.4	7.8	5.6	4.6	4.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.305	0.332	0.348	0.382	0.377	0.368	0.400	0.392	0.362	0.350	0.334	0.298	(40)
		taud	2.285	2.329	2.358	2.320	2.414	2.473	2.404	2.416	2.475	2.401	2.280	2.228	(41)
	Ebn at Noon		182	227	256	264	273	277	265	259	250	218	165	147	(42)
		Edh at Noon	16	23	29	35	34	32	34	31	26	22	16	13	(43)
(44) All-Sky Solar Radiation	RadAvg		113	302	781	1396	1866	2026	1831	1473	988	447	157	80	(44)
	RadStd		18	55	75	148	140	106	161	127	107	70	27	13	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (21 neighbors)	+0.70	N/A	+3.67	N/A	N/A	N/A	N/A	-234	+119	N/A
		+0.65	N/A	+2.16	N/A	N/A	N/A	-182	-252	N/A	N/A

Nomenclature: See separate page